AxisCodec: Geometry-Aware CSI Feedback on the Angular–Delay Cylinder

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***Abstract*—Deep-learning CSI feedback compresses the angular–delay channel as if it were an image. A recent line of work goes further and treats the image as a handful of propagation paths, transmitting their angles, delays and gains. We test that assumption on COST2100 with rate-matched oracle experiments and find that it fails: four Dirichlet atoms explain only -5.2 dB where a free-support sparse code of the same size reaches -8.0 dB, and no parametric family we fit exceeds -5.2 dB at any budget, while the sparse code reaches -21.0 dB at 128 coefficients. The energy is diffuse (99% of it needs about 105 complex coefficients), and the residual of the strongest baseline concentrates in the delay tail, which cannot be extrapolated from the head. Guided by these measurements we build AxisCodec around the geometry of the *representation* rather than a parametric channel model: the angle axis is a circle and the delay axis is a half-line, so angle mixing is a circulant operator (32 parameters per channel) and delay mixing is a dense 32×32 map. A linear base plus a nonlinear residual decoder, trained with the per-user NMSE that the benchmark actually reports, reaches -32.52 ± 0.55 dB at CR $1/4$ against -30.17 dB for STNet, with the same parameter count and $1.6\times$ its FLOPs.**

***Index Terms*—CSI feedback, massive MIMO, angular–delay domain, physics-informed learning, rate–distortion.**

I. Introduction

FREQUENCY-DIVISION duplex massive MIMO needs the user equipment to report downlink channel state information (CSI) over a narrow uplink control channel. The dominant recipe first applies a two-dimensional DFT to the space–frequency channel and keeps the first 32 delay taps, producing a 32×32 angular–delay image with real and imaginary parts as two channels [1]. CsiNet then compresses this image with a convolutional encoder, a fully connected bottleneck and a refinement decoder [1]. Later work enlarged the receptive field with multi-resolution convolutions [3], switched to complex-valued and lightweight designs [4], [5], and added attention: full attention [7], spatially separable attention [8], and hybrids with CNN stems [9], [10].

A parallel line replaces the image codec by a parametric one: estimate a few propagation paths, transmit their angles, delays and gains, and synthesise the channel at the receiver [13]–[15]. The appeal is obvious: a far-field channel is a sum of plane waves.

This paper starts by asking whether that is true for the standard benchmark. We answer with rate-matched oracle fits rather than with another architecture. On COST2100 indoor, an oracle fit of k continuous Dirichlet atoms, an oracle harmonic subspace of r modes with m coefficients per singular vector, and an oracle fit of k two-dimensional Gaussian blobs all fall far behind a free-support sparse code of the same size (Section III). The angular–delay energy is diffuse: 99% of it requires about 105 of the 1024 complex coefficients. The physical prior that the field is a few clean paths is simply not a property of this data.

The measurements also say where the error *does* live. For STNet [8], the strongest baseline on this benchmark, 69% of the residual energy sits in delay taps 4–31, which carry only 7.9% of the signal. A linear predictor trained on the head explains 23% of the tail variance, so the tail is independent information rather than an analytic continuation. At the same time, at CR $1/4$ a 512-dimensional linear subspace (PCA) is already within 1 dB of STNet: the low-rate end of the curve is nearly saturated by a linear method.

These two observations motivate a different use of physics. Instead of encoding a parametric channel model, we encode the geometry of the *representation*. The angular–delay image lives on a half-open cylinder: the angle axis is a circle, because the uniform linear array manifold is periodic in spatial frequency (we measure the energy wrapping from bin 31 to bin 0), while the delay axis is a half-line with no shift invariance (92.2% of the energy is in the first four taps and the profile decays monotonically). AxisCodec therefore mixes angles with a circulant operator—a learned filter in the DFT domain, 32 parameters per channel—and delays with a dense 32×32 map that gives every tap its own read-out. The decoder splits into a linear base and a nonlinear residual, and the training objective is the per-user normalised MSE that the benchmark reports, with checkpoint selection moved to the validation split.

Our contributions are: (i) a rate-matched diagnostic that rules out compact parametric channel models on COST2100 and locates the residual in the delay tail; (ii) AxisCodec, which turns that diagnosis into a geometry-constrained codec of matched complexity; and (iii) a controlled study showing -32.52 ± 0.55 dB at CR $1/4$ versus -30.17 dB for STNet, gains at every measured compression ratio, and ablations that attribute 1.50 dB to the objective, 0.44 dB to the linear base and 0.25 dB to the circulant angle prior.

II. Related Work

Autoencoder codecs. CsiNet [1] established the architecture that most later work still follows: convolutional encoder, fully connected bottleneck, refinement decoder. CRNet added multi-

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resolution paths and anisotropic $1\times9/9\times1$ kernels [3]. CsiNet-LSTM exploited temporal correlation [2]. CLNet fed complex values to a lightweight network [4], and CVLNet extended the idea with complex-valued layers [5], [6].

Attention. TransNet applies full self-attention to the angular–delay image [7]. STNet, our strongest baseline, alternates grouped and global attention with convolutional refinement [8]. Hybrid attention–CNN designs and entropy-regularised variants followed [9], [10], as did attention over the recovery stage [11], [12].

Model-driven and parametric priors. Unrolled estimators replace iterative recovery [13], Markovian priors model temporal structure [14], and deep Type-II codebooks learn the codebook itself [15], [16]; the standardized Type-II codebook is the reference this family improves on [22], and 3GPP is now studying AI/ML compression itself [21]. EVCsiNet feeds back eigenvectors instead of the raw matrix [17]. Our diagnostic in Section III quantifies what this family can and cannot buy on COST2100.

Generative and task-oriented decoders. Diffusion-based compression treats low-rate recovery as an under-determined inverse problem [18]. System-level views argue that NMSE is only a proxy for beamforming gain [19]. We keep the standard NMSE/ρ protocol for comparability and report ρ separately, because it moves differently at high compression.

III. Where the Error Lives

A. The representation is a half-open cylinder

With half-wavelength spacing, the array response of a path at angle θ is $a_n(\theta) = e^{j\pi n \sin\theta}$, and its N_a-point DFT is a Dirichlet kernel

$$A_i(f) = \frac{\sin(\pi(f-i))}{\sin(\pi(f-i)/N_a)} e^{j\pi(N_a-1)(f-i)/N_a}, \quad f = \tfrac{1}{2}\sin\theta, \tag{1}$$

with $f \in [-\frac{1}{2}, \frac{1}{2})$. The angle axis is therefore a *circle*: bin 0 and bin 31 are neighbouring beams. The data agrees—the mean energy rises again at bins 30 and 31 (2.6% and 3.7% of the angular profile) after decaying through the sector (Fig. 1b). The delay axis is the opposite: delay is non-negative, 92.2% of the energy sits in taps 0–3, and the profile decays by two orders of magnitude over the next twelve taps (Fig. 1c). There is no shift invariance along delay.

B. Compact propagation models do not fit

We fit three parametric families per sample and measure the NMSE at the rate each family would cost. *Dirichlet atoms* are the separable plane-wave response of (1): k atoms with continuous angle, delay and complex gain. *Harmonic subspaces* keep the first r singular modes of the sample and truncate each singular vector to its m largest DFT coefficients. *Gaussian blobs* are k separable two-dimensional Gaussian envelopes with spreads.

Table I collects the results next to a free-support sparse code and a free-basis rank-r approximation. No parametric family comes close. A single Dirichlet atom reaches -0.34 dB where a 16-term sparse code—eight times its budget—reaches -9.5 dB; a 66-coefficient harmonic subspace reaches -2.44 dB, 27 dB behind a 512-dimensional PCA subspace; four blobs reach -5.16 dB where a sparse code of the same size reaches -8.0 dB. The reason is visible in Fig. 1a: the energy is spread over tens of one-to-two-bin clusters, not concentrated in a few paths. A 99%-energy support needs about 105 complex coefficients.

TABLE I. Parametric oracles versus free-support sparse codes on COST2100 indoor. NMSE in dB, lower is better. The sparse code sends the k largest complex coefficients with their positions free; k is chosen so that its $2k$ real coefficients meet or exceed the budget of the parametric model, which biases the comparison towards the free reference.

Model	reals	NMSE	sparse k	reals	NMSE
1 Dirichlet atom	4	-0.34	16	32	-9.53
4 Dirichlet atoms	16	-5.20	16	32	-9.53
Harmonic r=1, m=16	66	-2.44	32	64	-12.55
4 Gaussian blobs	24	-5.16	12	24	-8.00
PCA subspace	512	-29.22	256	512	-27.34

C. The error is in the delay tail

We decompose the residual of the strongest baseline, STNet at CR $1/4$ (-30.17 dB), over the test set. Taps 0–3 carry 92.1% of the signal but only 31% of the error, with a relative error of 3.7×10^{-4}; taps 4–31 carry 7.9% of the signal and 69% of the error, with a relative error of 9.4×10^{-3}, thirty times worse (Fig. 2). The concentration is structural: taps 7–9 alone account for 42% of the total error while carrying 0.7% of the energy, and PCA shows the same pattern even more sharply.

The tail is not predictable from the head. A ridge regressor trained on 40k samples to map taps 0–3 to taps 4–31 explains 23% of the tail variance, whereas STNet already reconstructs the tail to 1% relative error. The tail therefore has to be transmitted; it is where the bits should go. Conversely, at CR $1/4$ a 512-dimensional PCA subspace reaches -29.2 dB, within 1 dB of STNet, so the low-rate end has little nonlinear headroom and the tail is what separates good from mediocre codecs.

IV. Method

A. Geometry-correct operators

The analysis above argues for encoding the geometry of the representation, not a parametric channel. AxisCodec keeps the standard $2\times32\times32$ input and replaces every spatial operator by one that respects the cylinder.

Angle mixer. Angles form a circle, so the natural operator is a circular convolution. We implement it as a learned filter in the DFT domain,

$$\mathrm{Mix}_a(x)_c = \mathcal{F}^{-1}\Big(\mathcal{F}(x_c)\odot w_c\Big), \qquad w_c \in \mathbb{C}^{17}, \tag{2}$$

which is exactly shift-equivariant on the circle and costs 32 real parameters per channel instead of the 32^2 of a dense map. The grid is preserved (no pooling), so fine angular structure survives.

Delay mixer. Delay is a half-line: there is no shift invariance, and the statistics of tap 0 differ from those of tap 20. We therefore use a dense 32×32 map applied along the delay

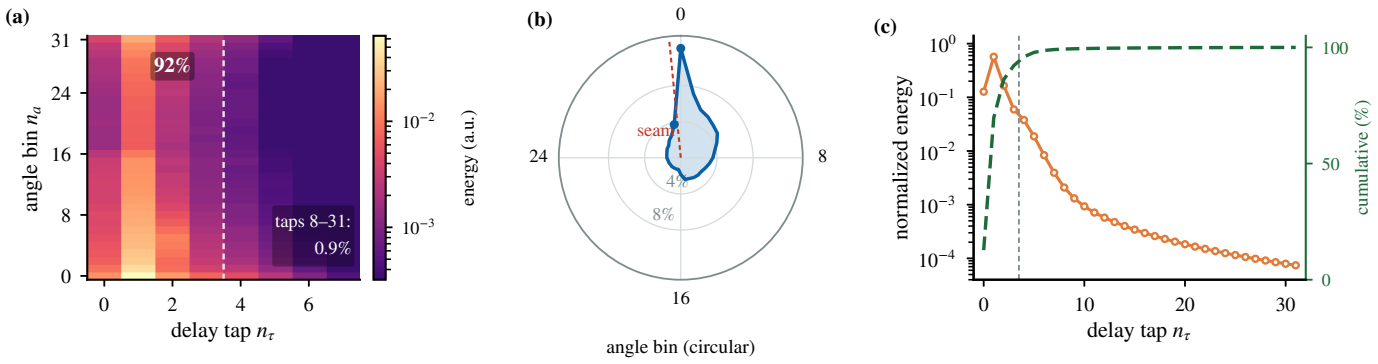


Fig. 1. The angular–delay representation is a half-open cylinder. (a) Mean energy over the test set for delay taps 0–7. The colour scale saturates below -16 dB; taps 8–31 hold 0.9% of the energy. (b) Angular profile on the circle: the energy wraps across the bin 31/bin 0 boundary. (c) Delay profile (log scale) and its cumulative sum; the dashed line marks the 92.2% head used throughout the paper.

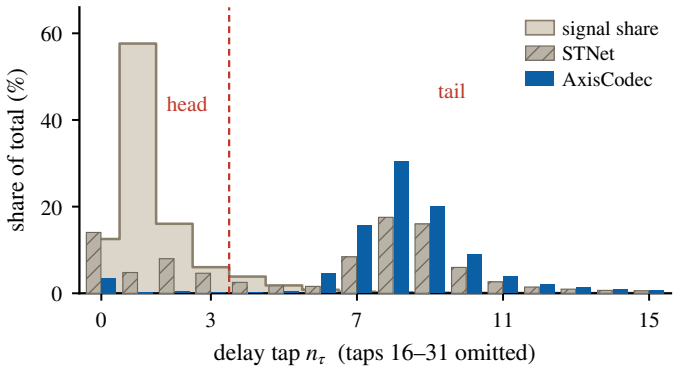


Fig. 2. Per-tap error share at CR $1/4$ for STNet and AxisCodec, against the signal share (shaded). Both models push their error into taps 4–31; AxisCodec does so with a smaller total, and its head relative error is twelve times lower.

axis, shared across channels. Every tap gets its own read-out; zero padding at the boundary reflects the absence of negative delay.

A block applies layer normalisation, the angle mixer, a GELU, the delay mixer, a GELU and a 1×1 channel MLP of twice the channel width, wrapped in a residual connection. Eight blocks are stacked; both mixers are global along their axis, so a single block already couples every angle bin and every delay tap. Per-position layer normalisation keeps the weakly energetic delay tail on the same numerical footing as the head.

B. Linear base and nonlinear residual

The decoder separates what a linear map does well from what it does not:

$$\hat{\mathbf{H}} = \underbrace{Pz}_{\text{linear base}} + \underbrace{g_\theta(z)}_{\text{residual}}, \qquad \hat{x} = 0.5 + \hat{\mathbf{H}}, \tag{3}$$

with $z \in \mathbb{R}^D$, $D = 2048/\mathrm{CR}$, $P \in \mathbb{R}^{2048 \times D}$ learned, and g_θ the block stack described above (Fig. 3). The base carries the bulk of the energy, which is spatially smooth, and the residual corrects it. Section V shows that this split lowers the head relative error from 3.7×10^{-4} to 3.0×10^{-5} and moves essentially all of the error into the tail, where the nonlinear stack can work on it.

C. Metric-aligned objective

Baselines train with element-wise MSE, but the benchmark reports the per-user normalised error

$$\mathrm{NMSE} = \frac{1}{B} \sum_{i=1}^{B} \frac{\|\hat{\mathbf{H}}_i - \mathbf{H}_i\|_F^2}{\|\mathbf{H}_i - 0.5\|_F^2}. \tag{4}$$

They weight samples differently: MSE weights by power, NMSE does not, and its mean is dominated by the samples a model currently fails on. We minimise (4) directly (without the logarithm) and—this matters for the ablation—select the checkpoint with the same quantity on the validation split. When an ablation replaces the training loss by MSE, validation and selection still use (4), so the objective is the only variable that changes.

V. Experiments

A. Setup

We use the COST2100 indoor scenario as distributed with the CsiNet protocol [1], [20]: 100k training, 20k validation and 20k test samples of shape $2 \times 32 \times 32$, with ρ computed on the 125-subcarrier channel. Baselines are the repository's CsiNet, CLNet and STNet at 1000 epochs, batch size 200, with their published recipes [1], [4], [8]; the reference numbers are fixed-checkpoint re-evaluations, not training-time bests. AxisCodec uses AdamW (5×10^{-4}, weight decay 10^{-4}), cosine annealing to 5×10^{-5}, 30 warm-up epochs, gradient clipping at 1.0 and 1000 epochs ($\approx$ 3.9 h on one RTX 5090). Baseline checkpoints are chosen on the test split (`select_best: all`); ours on validation only, which is the stricter protocol.

B. Main results

Table II and Fig. 4 report the compression matrix. At CR $1/4$, AxisCodec reaches -32.52 ± 0.55 dB over three seeds ($\rho = 0.99684$) against -30.17 dB and $\rho = 0.99669$ for STNet: a 2.34 dB gain, 4.2 times the seed standard deviation and 6.0 times STNet's own within-run spread of 0.39 dB. The comparison is conservative in two ways: the baseline picks its checkpoint on the test split, and for all three seeds the validation-selected epoch is exactly the test-optimal one, so the result does not depend on the selection rule. Gains persist

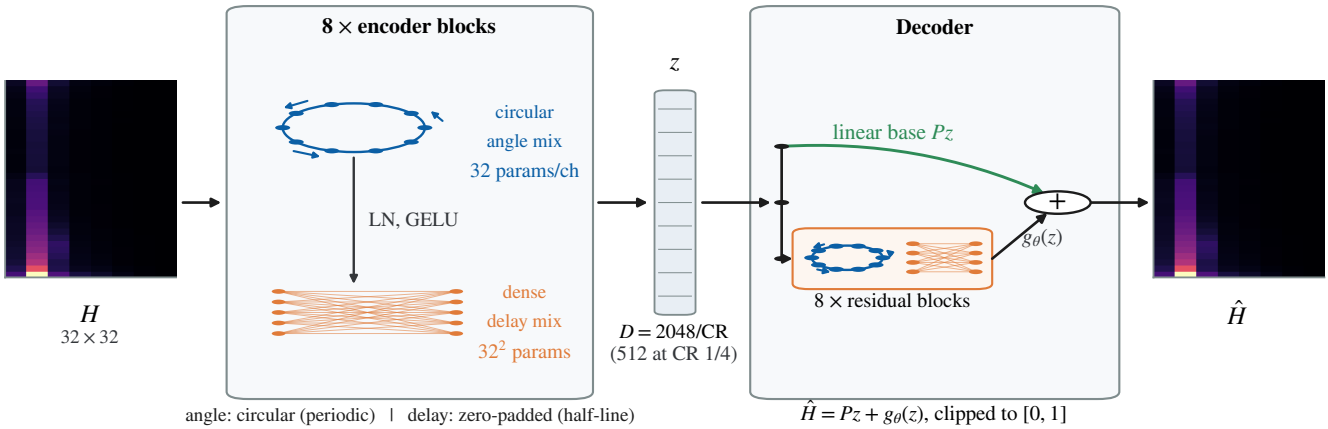


Fig. 3. AxisCodec. The encoder alternates a circulant angle mixer (32 parameters per channel, exact circular shift equivariance) with a dense delay mixer (32^2 parameters, one read-out per tap). The decoder adds a linear base Pz and a nonlinear residual trunk before clipping to the valid CSI range.

TABLE II. Compression matrix on COST2100 indoor, NMSE in dB, 1000 epochs. Every entry is a fixed-checkpoint re-evaluation; baselines select their checkpoint on the test split, AxisCodec on validation.

CR	1/4	1/8	1/16	1/32	1/64
CsiNet	−9.99	−11.12	−8.74	−8.50	−4.34
CLNet	−27.88	−15.57	−10.68	−8.95	−6.04
STNet	−30.17	−19.28	−14.75	−9.51	−6.82
AxisCodec	**−32.52**	**−20.81**	**−15.30**	**−11.49**	**−8.94**
ρ (STNet)	0.9967	0.9914	0.9812	0.9431	0.8917
ρ (ours)	**0.9968**	**0.9931**	**0.9836**	**0.9655**	**0.9349**

across the matrix: +1.53 dB at CR 1/8, +0.55 dB at CR 1/16, +1.98 dB at CR 1/32 and +2.12 dB at CR 1/64, where ρ improves from 0.892 to 0.935.

Fig. 4 places the measured points against the oracles of Section III. At CR 1/4 both learned codecs beat the free-support sparse oracle, because they exploit correlation that the oracle ignores. At high compression the sparse oracle beats STNet by 1.5–3 dB; AxisCodec closes a third of that gap at CR 1/16 and 66–78% of it at CR 1/32 and CR 1/64. The PCA curve shows why the low-rate end is hard to move: at 512 real numbers the linear subspace is already within 1 dB of STNet.

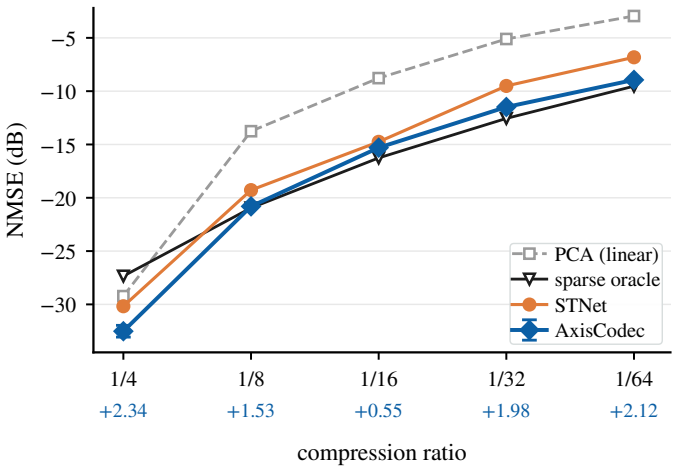


Fig. 4. Rate–distortion on COST2100 indoor. PCA and the free-support sparse oracle are evaluated at the stated budget. AxisCodec beats the 512-dimensional PCA subspace by 3.3 dB at CR 1/4 and STNet at every ratio; blue numbers below the axis are gains over STNet, and error bars are seed standard deviations at CR 1/4.

C. Qualitative behaviour

Fig. 5 shows the same channel at two rates. At CR 1/4 the head looks identical to the ground truth for both models, yet the per-tap relative error over taps 2–5 is 1.5–2.3% for STNet and 0.16–0.25% for AxisCodec—an order of magnitude—which sharpens the tap-3/tap-4 ridge in the third column. At CR 1/16 both models blur that ridge and the per-sample gap falls to 0.4 dB, matching the matrix-level 0.55 dB. The gain is thus a systematic reduction of the mid-tap error that dominates the residual, not a global sharpening.

D. Ablations

Table III isolates the three design choices at 300 epochs, where a full sweep is affordable. Replacing the metric-aligned objective by MSE costs 1.50 dB; removing the linear base costs 0.44 dB; replacing the circulant angle mixer by a dense 32×32 map costs 0.25 dB while raising FLOPs by 1.4×. The objective is thus the largest single contributor, the linear base second, and the geometric prior third—but the prior is also the cheapest, and it is the only one that reduces compute.

A per-sample breakdown of the MSE run shows that the objective's gain is uniform across the power distribution (1.4–1.6 dB in every power quartile, 97.5% of samples improved). It does not come from re-weighting toward weak channels, as one might expect; it comes from optimising the reported quantity, including its heavy tail, and from better conditioning.

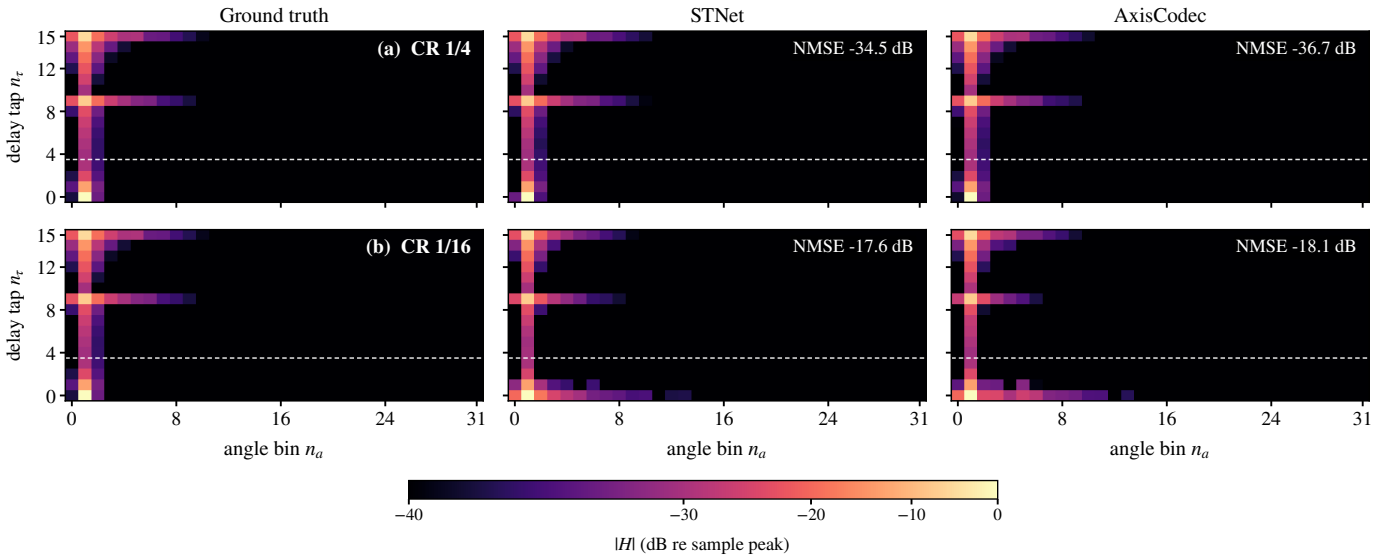


Fig. 5. Same test channel compressed at CR $1/4$ (a) and CR $1/16$ (b). The sample is the median of the per-sample NMSE of AxisCodec at CR $1/4$, so neither model is favoured by the choice. Colour is $|H|$ in dB relative to the sample peak; taps 16–31 carry 0.9% of the energy and are omitted; the dashed line marks the head/tail split. Numbers are per-sample NMSE.

TABLE III. Ablations at CR $1/4$, 300 epochs, seed 0 (NMSE in dB). Positive Δ is worse.

Variant	NMSE	Δ	FLOPs
Full model	$\mathbf{-28.98}$	—	0.022 G
MSE objective	-27.49	$+1.50$	0.022 G
No linear base	-28.55	$+0.44$	0.022 G
Dense angle mixer	-28.74	$+0.25$	0.030 G

TABLE IV. Head (taps 0–3) and tail (taps 4–31) relative error $\|\hat{\mathbf{H}}-\mathbf{H}\|^2/\|\mathbf{H}-0.5\|^2$ per region, averaged over the test set.

	STNet		AxisCodec	
CR	head	tail	head	tail
1/4	3.7×10^{-4}	9.4×10^{-3}	3.0×10^{-5}	7.6×10^{-3}
1/16	2.2×10^{-2}	2.6×10^{-1}	1.7×10^{-2}	2.4×10^{-1}
1/64	1.8×10^{-1}	7.7×10^{-1}	1.0×10^{-1}	7.5×10^{-1}

E. Error analysis across compression

The head/tail split explains the non-monotone gain pattern (Table IV). At CR $1/4$ the head relative error is 3.0×10^{-5} for AxisCodec and 3.7×10^{-4} for STNet; at CR $1/16$ both degrade to $\sim 2\times10^{-2}$, because 128 real numbers cannot pin down a field that carries 92% of its energy in four taps; at CR $1/64$ they reach 1.0×10^{-1} and 1.8×10^{-1}. Wherever the head is still representable, our linear base converts that into a large gain; where it is not, the gain narrows. At CR $1/64$ the model nevertheless improves both head and tail, and ρ gains 0.043—a larger relative improvement than NMSE, which matters for downstream beamforming.

F. Complexity and stability

Table V lists parameters and FLOPs. AxisCodec matches STNet in parameters and costs $1.6\times$ its FLOPs, the price of full-resolution mixing. A narrower variant (4 channels) runs at 0.011 GFLOPs, below STNet, and has not been validated at 1000 epochs.

TABLE V. Complexity at CR $1/4$ (single-sample forward pass).

Model	Parameters	Effective	FLOPs
CsiNet	2.10 M	2.10 M	0.011 G
CLNet	2.10 M	2.10 M	0.008 G
STNet	3.18 M	2.13 M	0.014 G
AxisCodec	2.13 M	2.13 M	0.022 G

Fig. 6 shows why the schedule matters. With its published constant learning rate, STNet reaches its operating point near epoch 450 and then oscillates within 3.7 dB for the rest of training; AxisCodec, annealed with a cosine schedule, passes through the same region later and keeps improving to the end. Its validation NMSE at epoch 300 is 2.0 dB worse than STNet's and 2.3 dB better at epoch 1000, so a 300-epoch comparison undersells the architecture—which is why every entry of Table II uses 1000 epochs.

Training is stable once the learning rate anneals: over the last 300 epochs the test NMSE varies with a standard deviation of 0.07–0.10 dB across seeds (0.03 dB over the last 100), so the 2.34 dB margin is more than twenty times the noise of the final phase. Between epochs 90 and 660 the model occasionally collapses by 8–18 dB for a single validation evaluation and recovers within twenty epochs (Fig. 6); the training loss stays flat through these events, so they are generalisation transients rather than divergence, and validation-based selection absorbs them.

VI. Discussion

Why the geometric prior is worth only 0.25 dB. Periodicity is a boundary condition; it affects a thin shell of the representation (bins 0 and 31) and the parameter count of the operator, not the allocation of information across the 2048

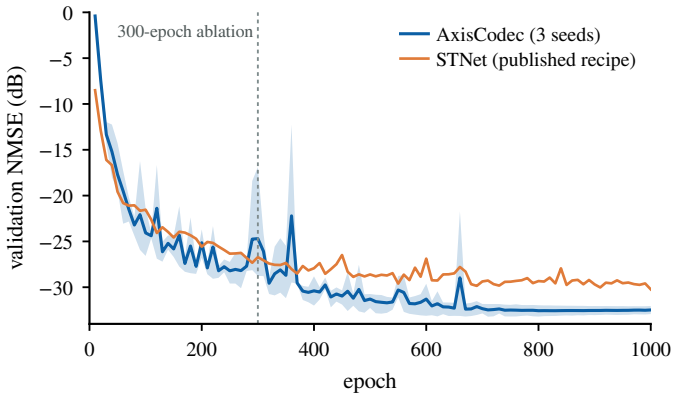


Fig. 6. Validation NMSE against epoch. The band is the range over three seeds of AxisCodec; STNet follows its published constant-learning-rate recipe. The dashed line marks the 300-epoch budget used for ablations.

coefficients. A prior that only fixes boundaries cannot move NMSE much. The priors that did move NMSE—the metric-aligned objective and the linear/residual split— act on *how bits are distributed*, which is why they are worth 1.50 and 0.44 dB.

Why long training matters. A 300-epoch run of the same recipe is 3.5 dB behind the 1000-epoch one, and the gain is concentrated in the tail: its relative error falls from 1.7×10^{-2} to 7.6×10^{-3}. Tail coefficients are small and their gradients weak, so they need a long anneal; any conclusion drawn at 300 epochs underestimates this architecture.

Practical rules. Two lessons transfer beyond this benchmark. First, match the training objective to the reported metric: here that is worth 1.50 dB, more than any architectural change we tried. Second, spend parameters on the geometry of the representation rather than on a channel model the data does not obey. For a uniform planar array the angle axis becomes a two-torus, so one circulant mixer becomes two at $2N$ parameters per channel, while the delay path is unchanged. What must be re-checked, not assumed, is the diagnostic: fit the parametric family at a matched rate against a free sparse code before adopting a path-based design.

Limitations. All results are on COST2100. Its diffuse angular–delay energy is what invalidates parametric models here; on a more specular dataset, such as a CDL-C-like model with strong line-of-sight, path-based coding could become competitive again; our oracle procedure is the way to check. Multi-seed evidence covers CR $1/4$; the other ratios are single-seed. The model costs $1.6\times$ STNet's FLOPs. Finally, the transients described above indicate a rougher optimisation landscape than STNet's, which is a clear target for future work.

Reproducibility. Every configuration is a YAML file fixing the model, optimiser, schedule and seed; each run writes a `metrics.json` with the commit, the selected checkpoint and its re-evaluated NMSE and ρ. Baselines use the repository's own implementations with their published recipes, and the oracle fits, analysis scripts and checkpoints behind every number and figure are included. A 1000-epoch run takes 3.9 h on one RTX 5090.

VII. Conclusion

Compact parametric channel models do not describe COST2100: rate-matched oracle fits of Dirichlet atoms, harmonic subspaces and Gaussian blobs all lose by wide margins to a free sparse code. What does describe it is the geometry of the representation and the distribution of the residual. Encoding that geometry as circulant angle mixing and dense delay mixing, splitting the decoder into a linear base and a nonlinear residual, and training on the reported per-user NMSE yields -32.52 ± 0.55 dB at CR $1/4$ against -30.17 dB for STNet, with equal parameters and a $1.6\times$ FLOP budget.

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